5. Weddle's rule (Numerical integration) example (Enter your problem)

	Other related methods
1. Formula & Example-1 (table data) 2. Example-2 ($f(x) = \frac{1}{x}$)	1. Trapezoidal rule 2. Simpson's 1/3 rule 3. Simpson's 3/8 rule 4. Boole's rule 5. Weddle's rule

4. Boole's rule
(Previous method)

2. Example-2 ($f(x) = \frac{1}{x}$)
(Next example)

1. Formula & Example-1 (table data)

Formula

1. Weddle's Rule

$$\int y\,dx = \frac{3h}{10}\left[\left(y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + y_6\right) + \left(y_6 + 5y_7 + y_8 + 6y_9 + y_{10} + 5y_{11} + y_{12}\right) + \ldots\right]$$

Examples

1. Find Solution using Weddle's rule

x	f(x)
7.47	1.93
7.48	1.95
7.49	1.98
7.50	2.01
7.51	2.03
7.52	2.06
7.53	2.09

Solution:

Question 2: Write a python code of Newton Raphson method and display its output

```
def f(x):
    return x**3 - x - 2

def df(x):
    return 3*x**2 - 1

def newton_raphson(x0, tol, max_iter):
    for i in range(max_iter):
        x1 = x0 - f(x0)/df(x0)
        print(f"Iteration {i+1}: x = {x1:.6f}")
        if abs(x1 - x0) < tol:
            return x1
        x0 = x1
    return x0

root = newton_raphson(1.5, 0.0001, 10)
print(f"\nRoot using Newton-Raphson Method: {root:.6f}")
```

```
Iteration 1: x = 1.521739
Iteration 2: x = 1.521380
Iteration 3: x = 1.521380

Root using Newton-Raphson Method: 1.521380
```

Question 3: Write a python code of Iteration method and display its output

```
def g(x):
    return (x + 2 / x**2) / 2  # From x^3 = x + 2

def fixed_point(x0, tol, max_iter):
    for i in range(max_iter):
        x1 = g(x0)
        print(f"Iteration {i+1}: x = {x1:.6f}")
        if abs(x1 - x0) < tol:
            return x1
        x0 = x1
    return x0

root = fixed_point(1.5, 0.0001, 20)
print(f"\nRoot using Iteration Method: {root:.6f}")
```

```
Iteration 1: x = 1.194444
Iteration 2: x = 1.298142
Iteration 3: x = 1.242482
Iteration 4: x = 1.269009
Iteration 5: x = 1.255474
Iteration 6: x = 1.262168
Iteration 7: x = 1.258804
Iteration 8: x = 1.260481
Iteration 9: x = 1.259641
Iteration 10: x = 1.260061
Iteration 11: x = 1.259851
Iteration 12: x = 1.259956
Iteration 13: x = 1.259904

Root using Iteration Method: 1.259904
```

The value of table for x and y

x	7.47	7.48	7.49	7.5	7.51	7.52	7.53
y	1.93	1.95	1.98	2.01	2.03	2.06	2.09

Using Weddle's Rule

$$\int y dx = \frac{3h}{10}\left[y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + y_6\right]$$

$$\int y dx = \frac{3 \times 0.01}{10}[1.93 + 5 \times 1.95 + 1.98 + 6 \times 2.01 + 2.03 + 5 \times 2.06 + 2.09]$$

$$\int y dx = 0.1204$$

Solution by Weddle's Rule is 0.1204

2. Find Solution using Weddle's rule

x	f(x)
0	1
1	0.5
2	0.2
3	0.1
4	0.0588
5	0.0385
6	0.027

Solution:
The value of table for x and y

x	0	1	2	3	4	5	6
y	1	0.5	0.2	0.1	0.0588	0.0385	0.027

Using Weddle's Rule

$$\int y dx = \frac{3h}{10}\left[y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + y_6\right]$$

$$\int y dx = \frac{3 \times 1}{10}[1 + 5 \times 0.5 + 0.2 + 6 \times 0.1 + 0.0588 + 5 \times 0.0385 + 0.027]$$

$$\int y dx = 1.37349$$

Solution by Weddle's Rule is 1.37349

This material is intended as a summary. Use your textbook for detail explanation.
Any bug, improvement, feedback then Submit Here

4. Boole's rule
(Previous method)

2. Example-2 ($f(x) = \frac{1}{x}$)
(Next example)

4. **Iteration 4:**
   - Interval: $[-3.25, -3.125]$
   - Midpoint: $c = \frac{-3.25+(-3.125)}{2} = -3.1875$
   - $f(-3.1875) = 2e^{-3.1875} - \sin(-3.1875) \approx 0.083 - (-0.047) \approx 0.130$ (Positive)
   - New interval: $[-3.25, -3.1875]$
5. **Iteration 5:**
   - Interval: $[-3.25, -3.1875]$
   - Midpoint: $c = \frac{-3.25+(-3.1875)}{2} = -3.21875$
   - $f(-3.21875) = 2e^{-3.21875} - \sin(-3.21875) \approx 0.080 - (-0.078) \approx 0.158$ (Positive)
   - New interval: $[-3.25, -3.21875]$

Final Approximation:

The root lies in $[-3.25, -3.21875]$. Taking the midpoint:

$$\text{Root} \approx \frac{-3.25 + (-3.21875)}{2} = -3.234375$$

Question No. 2

(a) Use Newton Raphson Method to find the root of the equation:

$$x^3 - 2x - 5 = 0$$

use the initial guess of $x_o = 2.5$ **[up to 3rd iteration]**

Solution:

Define the function and its derivative:

$$f(x) = x^3 - 2x - 5$$
$$f'(x) = 3x^2 - 2$$

The Newton-Raphson formula is:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Iterations:

1. **Iteration 1 ($x_0 = 2.5$):**

$$f(2.5) = (2.5)^3 - 2(2.5) - 5 = 5.625$$
$$f'(2.5) = 3(2.5)^2 - 2 = 16.75$$
$$x_1 = 2.5 - \frac{5.625}{16.75} \approx 2.1642$$

2. **Iteration 2 ($x_1 = 2.1642$):**

$$f(2.1642) \approx (2.1642)^3 - 2(2.1642) - 5 \approx 0.8106$$
$$f'(2.1642) \approx 3(2.1642)^2 - 2 \approx 12.0514$$
$$x_2 = 2.1642 - \frac{0.8106}{12.0514} \approx 2.0970$$

3. **Iteration 3 ($x_2 = 2.0970$):**

$$f(2.0970) \approx (2.0970)^3 - 2(2.0970) - 5 \approx 0.0265$$
$$f'(2.0970) \approx 3(2.0970)^2 - 2 \approx 11.191$$
$$x_3 = 2.0970 - \frac{0.0265}{11.191} \approx 2.0946$$

Final Approximation:

After 3 iterations, the approximate root is:

$$\text{Root} \approx 2.0946$$